

Novel Hybrid Gate Ferroelectric Transistor-based Weight Device with High Linearity and Symmetry for On-Chip Learning

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Abstract

In this work, a novel hybrid gate ferroelectric (FE) transistor (HG-FeFET) based weight device is proposed and experimentally demonstrated for the first time, enabling high linearity and symmetry for enhanced on-chip learning. By leveraging the FE-thickness-dependent coercive voltage (V_C) distributions of FE layers, and the superposition effect of V_C distributions in parallel FE layers, the HG-FeFET, featuring hybrid FE layers of varying thickness connected in parallel, achieves a broadened V_C distribution. This results in a 3.6-fold improvement in linearity and symmetry under identical programming pulses, as well as higher weight precision. Based on the proposed HG-FeFET, the on-chip learning process of image recognition is demonstrated with a high accuracy of 95.33%, showing its great potential for edge AI applications. (Keywords: Ferroelectric FET, computing-in-memory, on-chip learning, linearity and symmetry)

Introduction

Deep neural networks (DNNs) have been widely adopted for edge applications such as embodied intelligence and autonomous driving (Fig. 1a), where on-chip learning is crucial for real-time adaptation to diverse environments [1-2]. Computing-in-memory (CIM) architectures based on high-density non-volatile memory (NVM), capable of performing highly parallel vector-matrix multiplication (VMM) and in-situ weight updates within memory arrays, have garnered significant attention for accelerating on-chip learning in DNNs [3]. Hafnium-based ferroelectric (FE) field-effect transistors (FeFETs) are particularly promising for CIM due to their high CMOS compatibility, low energy consumption, and high on/off ratios [4]. To enhance the accuracy of on-chip learning in FeFET-based CIM, improving the linearity and symmetry of FeFET weight updates is essential. Recent advancements include superlattice gate stack-based FeFETs with improved FE domain density [5-6] and 2T-1FeFET weight cells with hybrid precision [7], both of which aim to address linearity and symmetry challenges. However, these designs often rely on complex peripheral circuits for non-identical programming pulse schemes (Fig. 1b)

or intricate weight cell architectures, which still suffer from high hardware cost.

In this work, a novel ferroelectric transistor with a hybrid gate stack (HG-FeFET) incorporating FE layers of varying thicknesses is proposed as a weight device for on-chip learning (Fig. 2). By leveraging the broadened coercive voltage (V_C) distribution of parallel FE layers, the HG-FeFET is experimentally demonstrated to achieve improvement in linearity and symmetry, along with more conductance states within a single device. Based on this design, the high-accuracy on-chip learning of DNN is achieved, highlighting its great potential for edge applications.

Characteristics of ferroelectric capacitors with hybrid layers of differing thicknesses

The FeFET leverages non-volatile FE polarization in the FE gate to modulate channel conductance. The linearity and symmetry of the weight update process are directly influenced by the uniformity of polarization changes in the FE layer in response to programming pulses [5-6]. The structure of the proposed HG-FeFET weight device is illustrated in Fig. 2a. The hybrid gate stack consists of two FE layers with varying thicknesses connected in parallel, designed to broaden the overall distribution of ferroelectric coercive voltage (V_C) and improve programming uniformity. To analyze this behavior, we first investigate the properties of FE layers with different thicknesses using fabricated ferroelectric capacitors (FeCaps).

A. FeCap fabrication and modeling

The FeCaps of $\text{TiN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2/\text{TiN}$ structure (Fig. 3a) with varying FE layer thicknesses are experimentally fabricated and analyzed based on the multi-domain Preisach FE model [8] (Fig. 3b). In this model, the overall distribution of coercive field (E_C) for a large number of FE domains in the FE layer is assumed to follow a Gaussian distribution, which corresponds to the coercive voltage (V_C) distribution of the FeCaps. The V_C distribution and remanent polarization (P_r) of FeCaps with different FE layers are calibrated using measured P-V characteristics (Fig. 3c). The mean and variance of V_C distribution increase with the thickness of the FE layer, and the distribution is relatively concentrated (Fig. 4a). This

concentration leads to non-uniform polarization switching under continuous programming pulses in FeFETs with a constant FE layer thickness.

B. Behavior of FeCaps with hybrid thicknesses

In this work, we propose a novel hybrid FE layer design with varying thicknesses connected in parallel to broaden the V_C distribution through the superposition effect of independent distributions. As shown in Fig. 4b, the overall V_C distribution of the hybrid FeCap, simulated using the multi-domain Preisach FE model, appears more dispersed compared to a single FeCap. Moreover, due to the thickness-dependent P_r , the V_C distribution of hybrid FeCaps can be further optimized by adjusting the area ratio of the parallel-connected FeCaps with different thicknesses (Fig. 4c). Experimentally, the hybrid FeCap is constructed by connecting FeCaps with 10 nm and 15 nm FE layers in parallel. As shown in Fig. 5, the dynamic I-V curves confirm that the hybrid FeCap broadens the V_C distribution, leading to improved linearity and symmetry of FE polarization switching under continuous identical pulses (Fig. 6).

HG-FeFET based weight update behavior

The proposed HG-FeFET and conventional FeFET based weight cells are experimentally constructed by respectively connecting the hybrid FeCap and single-thickness FeCap to the gate of a MOSFET (Fig. 7ab). The I_d - V_g curves under relatively high sweep gate voltage exhibit typical FE hysteresis due to the polarization switching (Fig. 7c), indicating the weight storage capability. As shown in Fig. 8a-c, the continuous positive and negative programming pulses are applied to the gate of weight cells. The channel conductance (G_{DS}) is measured at the interval of pulses, corresponding to the long-term potentiation and depression behavior. Benefitting from the broadened V_C distribution, the HG-FeFET weight update exhibits significantly improved linearity and symmetry compared to the conventional FeFET, highlighting its potential as a weight device for on-chip learning.

Moreover, the HG-FeFET with optimized area ratio of FE layers with different thicknesses is investigated by HSPICE model, which is established based on the multi-domain Preisach FE model and BSIM-4 MOSFET model (Fig. 9a). As shown in Fig. 9b, under identical programming pulses, the proposed

HG-FeFET with a 2:1 area ratio of 15 nm and 8 nm FE layers achieves superior linearity and symmetry, showing a 3.6-fold improvement compared to conventional FeFET weight devices. Using the proposed HG-FeFET weight cell, the on-chip learning process of a two-layer neural network is demonstrated for image recognition (Fig. 10a). The network achieves an accuracy of 95.33% after 200 iterations (Fig. 10b), matching the precision expected for an ideal 6-bit weight. These results indicate that the proposed HG-FeFET weight device holds significant promise for edge on-chip learning in DNNs.

Conclusion

In this work, a novel HG-FeFET weight device is proposed and experimentally demonstrated with broadened VC distribution, enabling a significant improvement in linearity and symmetry for weight updates, as well as an increased number of intermediate conductance states. Based on the proposed HG-FeFET, on-chip learning is successfully demonstrated for image recognition with high accuracy, highlighting its great potential for edge AI applications.

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References

- [1] W. Zhang et al., "Edge learning using a fully integrated neuro-inspired memristor chip," *Science*, vol. 381, no. 6663, pp. 1205–1211, Sep. 2023, doi: [10.1126/science.ade3483](https://doi.org/10.1126/science.ade3483).
- [2] Z. Wang et al., "CPE: An Energy-Efficient Edge-Device Training with Multi-dimensional Compression Mechanism," in 2023 60th ACM/IEEE DAC, Jul. 2023, pp. 1–6, doi: [10.1109/DAC56929.2023.10247968](https://doi.org/10.1109/DAC56929.2023.10247968).
- [3] J. An et al., "Design memristor-based computing-in-memory for AI accelerators considering the interplay between devices, circuits, and system," *Science China Information Sciences*, vol. 66, no. 8, pp. 182404, Jul. 2023, doi: [10.1007/s11432-022-3627-8](https://doi.org/10.1007/s11432-022-3627-8).
- [4] Z. Zhang et al., "Recent progress of hafnium oxide-based ferroelectric devices for advanced circuit applications," *Science China Information Sciences*, vol. 66, no. 10, pp. 200405, Sep. 2023, doi: [10.1007/s11432-023-3780-7](https://doi.org/10.1007/s11432-023-3780-7).
- [5] K. A. Aabrar et al., "BEOL Compatible Superlattice FerroFET-based High Precision Analog Weight Cell with Superior Linearity and Symmetry," in 2021 IEEE IEDM, Dec. 2021, pp. 19.6.1–19.6.4, doi: [10.1109/IEDM19574.2021.9720713](https://doi.org/10.1109/IEDM19574.2021.9720713).
- [6] Y. Zhou et al., "Hybrid-FE-Layer FeFET With High Linearity and Endurance Toward On-Chip CIM by Array Demonstration," *IEEE Electron Device Letters*, vol. 45, no. 2, pp. 276–279, Feb. 2024, doi: [10.1109/LED.2023.3346030](https://doi.org/10.1109/LED.2023.3346030).
- [7] X. Sun et al., "Exploiting Hybrid Precision for Training and Inference: A 2T-1FeFET Based Analog Synaptic Weight Cell," in 2018 IEEE IEDM, Dec. 2018, pp. 3.1.1–3.1.4, doi: [10.1109/IEDM.2018.8614611](https://doi.org/10.1109/IEDM.2018.8614611).
- [8] J. Luo et al., "Capacitor-less Stochastic Leaky-FeFET Neuron of Both Excitatory and Inhibitory Connections for SNN with Reduced Hardware Cost," in 2019 IEEE IEDM, Dec. 2019, pp. 6.4.1–6.4.4, doi: [10.1109/IEDM19573.2019.8993535](https://doi.org/10.1109/IEDM19573.2019.8993535).

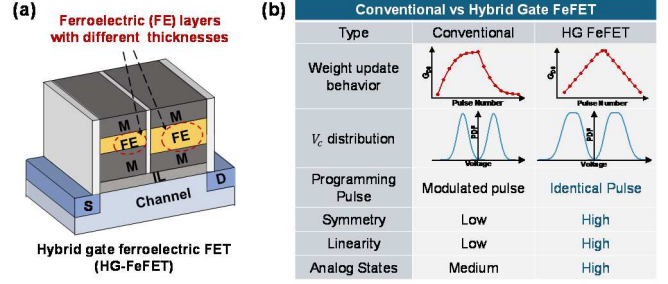
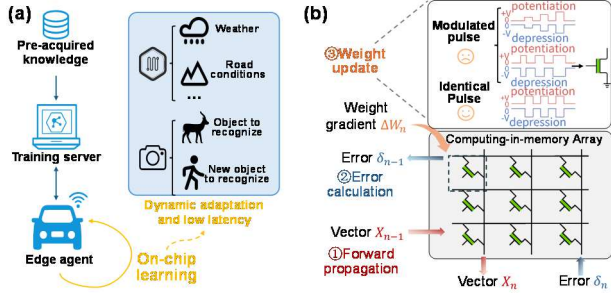


Fig. 1 (a) On-chip learning applications in autonomous driving. (b) On-chip learning process based on CIM array, and two programming pulses for weight update.

Fig. 2 (a) Schematic of hybrid gate ferroelectric FET. (b) Comparison of conventional ferroelectric and hybrid gate FeFET.

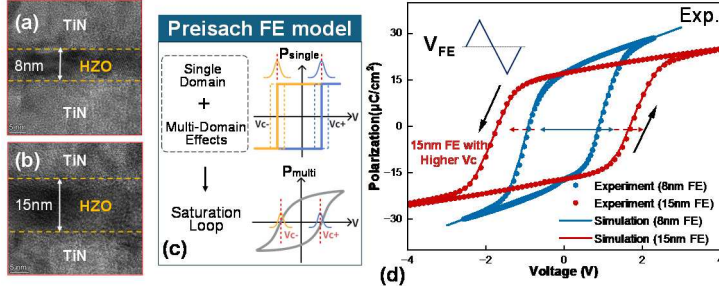


Fig. 3 (a) (b) TEM image of FE layer. (c) Multi-domain FE model based on the Preisach. (d) Measured P-V characteristics of the FeCAPs compared with the curve fitted using the Preisach model.

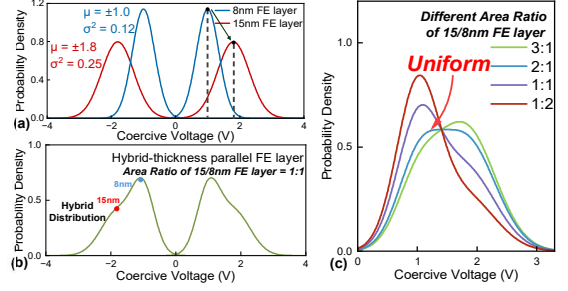


Fig. 4 Probability density functions of the V_c distributions for (a) 8nm and 10nm FE, (b) a hybrid FE layer, (c) hybrid FE with varying area ratios.

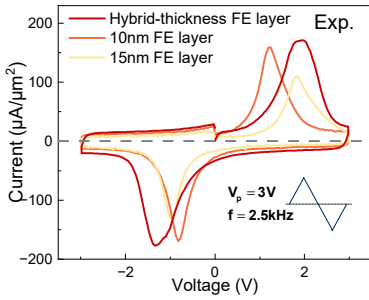


Fig. 5 Measured I-V characteristics showing stretched polarization current peaks of hybrid FE layer.

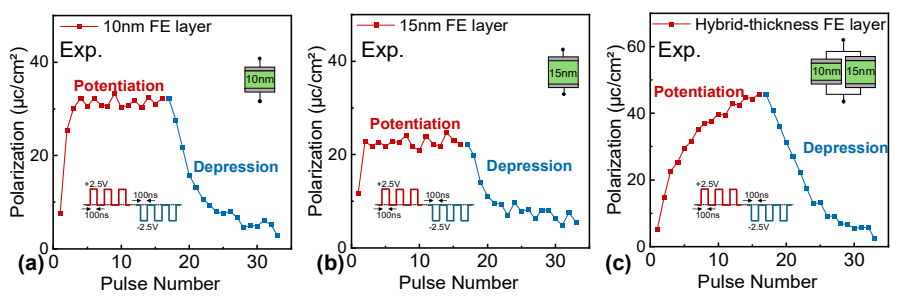


Fig. 6 The polarization versus pulse number diagrams for (a) 10nm single-thickness FE capacitor, (b) 15nm single-thickness FE capacitor, and (c) Hybrid-thickness FE capacitor (10nm and 15nm FE connected in parallel).

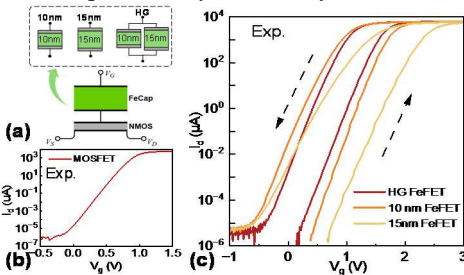


Fig. 7 (a) Experimental design for Hybrid Gate FeFET. (b) (c) Measured I-V characteristics of the MOSFET and FeFETs.

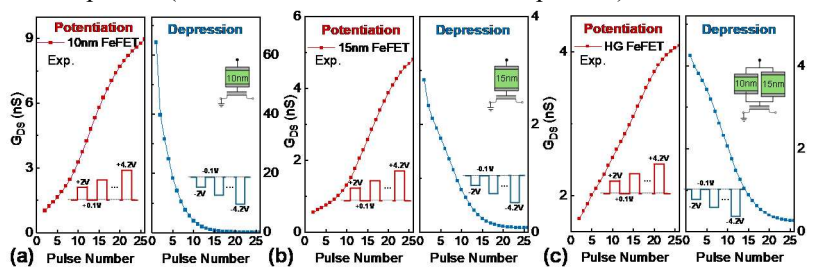


Fig. 8 Conductance versus pulse number diagrams for (a) 10nm single-thickness FeFET, (b) 15nm single-thickness FeFET, and (c) mixed-thickness FeFET (10nm and 15nm FE connected in parallel).

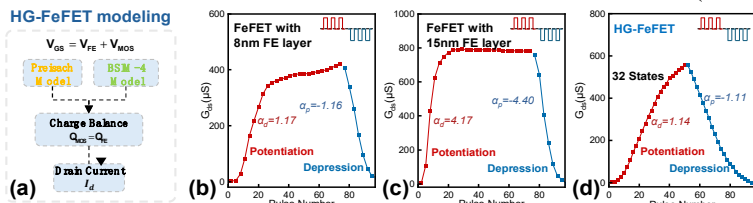


Fig. 9 (a) The modeling framework of HG-FeFET. (b) Simulated Conductance versus pulse number diagrams of FeFETs under identical pulse.

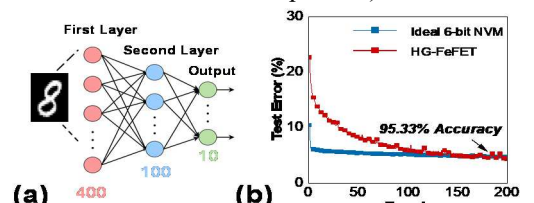


Fig. 10 (a) Diagram of neural network architecture for image recognition. (b) Simulated on-chip learning.